



What will be in the next release, **CF 1.14**

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A (very) brief history of CF releases

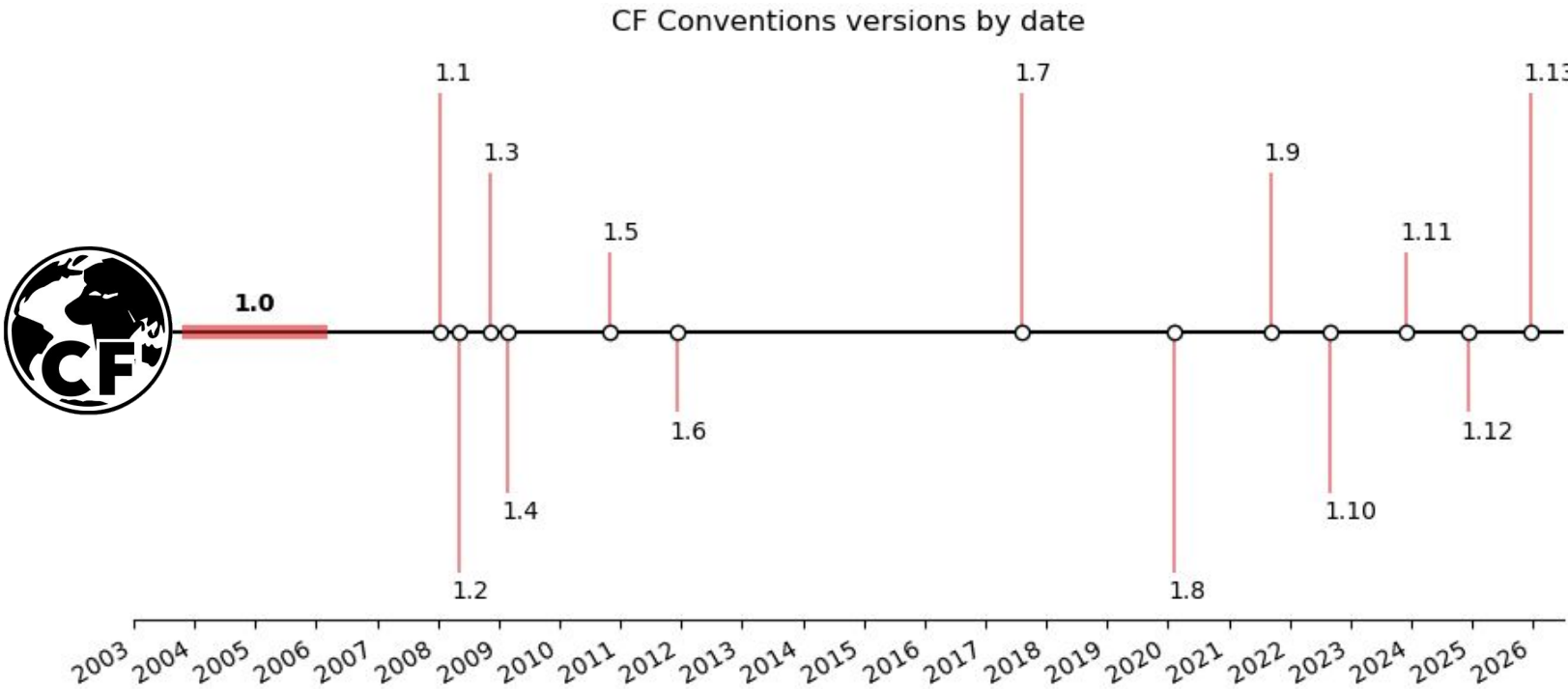
Where are we at? What's our version strategy?



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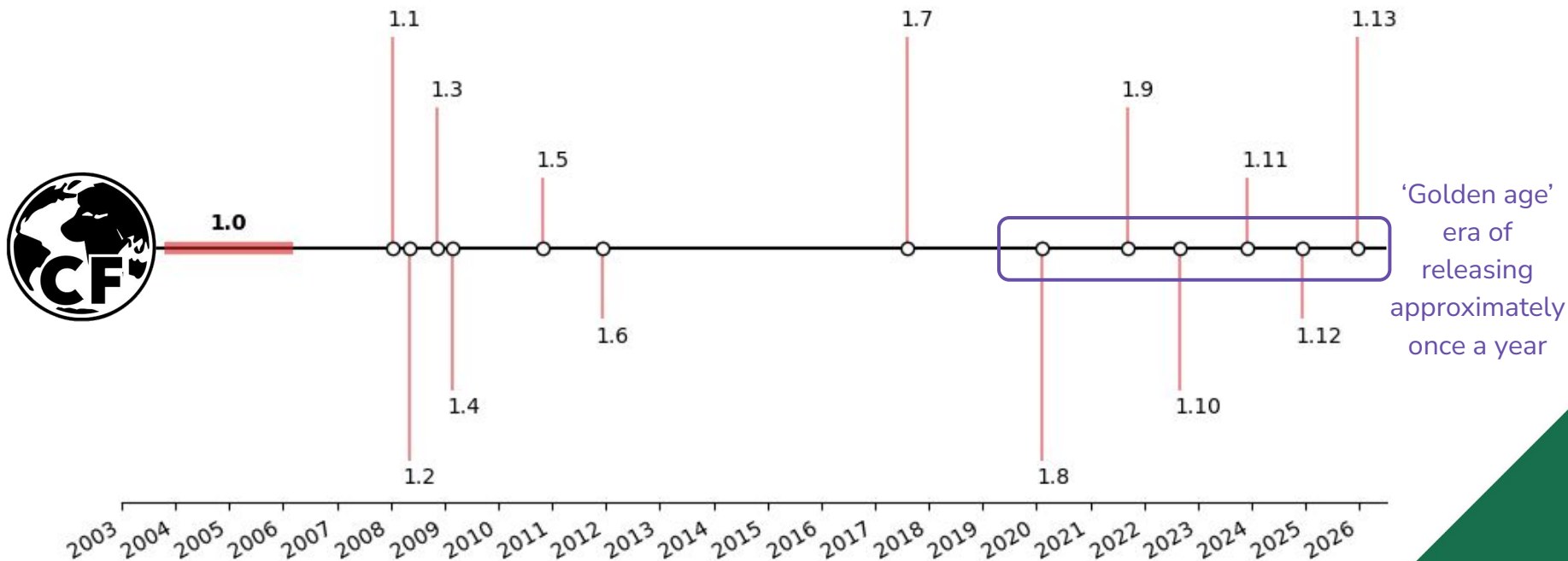
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A timeline of CF (canonical document) versioning*



A timeline of CF (canonical document) versioning*

CF Conventions versions by date



*Note **the controlled vocabularies have their own independent versioning**, mostly since the standard names are released much more often than the Conventions document itself (with current Standard Name Table version v94, released early June 2026)



CF versioning strategy

- **As of 2020**, releases have been every year (quite erratic pattern before then as per timeline of previous slide!)
- **As of 2023**: yearly release date scheduled a *short time after* (previously was *just before*) the annual workshop, to allow items decided on at the meeting to make it more rapidly into the published conventions” on a date agreed during the workshop
- **The idea**: regular releases whereby we can discuss and aim to progress and ideally conclude/merge unresolved loose ends during each yearly workshop
- For example, last year’s workshop was in mid/late September with the release mid December 2025 - so about 3 months later - aiming for similar this year for 1.14



Latest CF Conventions release: CF v1.14, due out in a few months

Replacing current latest CF v1.13 (Dec. 2025)

So what's changing?



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Definitely in (i.e. already merged into) CF 1.14

New items are listed in the 'Revision History' at the very foot of the (draft version) canonical conventions document, in this case for v1.14 at: https://cfconventions.org/cf-conventions/cf-conventions.html#_version_1_14_draft, or you can consult the GitHub milestone for any (made or upcoming) release, for v1.14: <https://github.com/cf-convention/cf-conventions/milestone/9>

- **Reduced Gaussian grid** support [\[#619\]](#)
 - no existing CF grid mapping can capture the characteristics of the reduced Gaussian grid, so a new convention and parameters are adopted to encode it within CF
- A new parametric vertical coordinate, '**atmosphere hybrid sigma log-pressure coordinate**' [\[#647\]](#)
 - Support for a vertical coordinate type used by the GEM atmospheric general circulation model. See new drafted sub-section under 'Appendix D: Parametric Vertical Coordinates'.
- **Clarify units prohibition** of numerical factors and offsets [\[#620\]](#)
 - CF explicitly imposes a restriction on a subset of the syntax that UDUNITS permits
- Section '**4.3. Vertical (Height or Depth) Coordinate**' clarity improvements [\[#635\]](#)
- **Formatting/consistency and infrastructure** improvements and some "bells and whistles":
 - Consistently reference "CF conventions" not "standard" or just "convention", and use the impersonal form ("we recommend" → "it is recommended") [\[#623\]](#)
 - Consistently represent datetimes and use calendar attribute in examples [\[#641\]](#)
 - Configurable dark mode for HTML view [\[#607\]](#)
 - Updated website e.g. homepage improvements (not technically tagged with v1.14 but new since last year) [e.g. cf-convention/cf-convention.github.io#605]



Some definite inclusions, illustrated

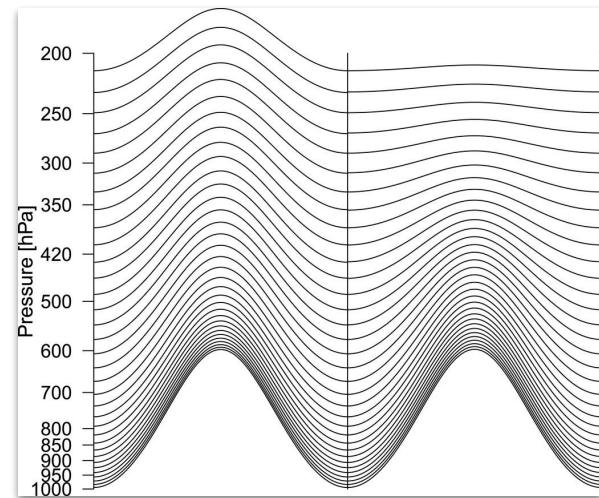
Atmosphere hybrid sigma

log-pressure coordinate

Rectification coefficients control how rapidly the vertical levels are rectified over the mountain.

“Vertical model-level structure showing ζ levels as functions of pressure for values below 200 hPa ...in a typical operational configuration [for an illustrative] sinusoidal mountain” from

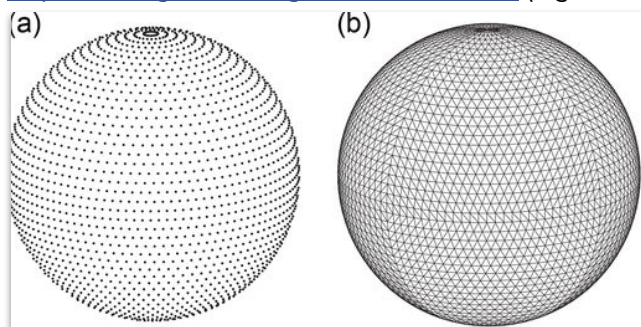
<https://doi.org/10.1175/MWR-D-13-00255.1>:



Reduced Gaussian grid support

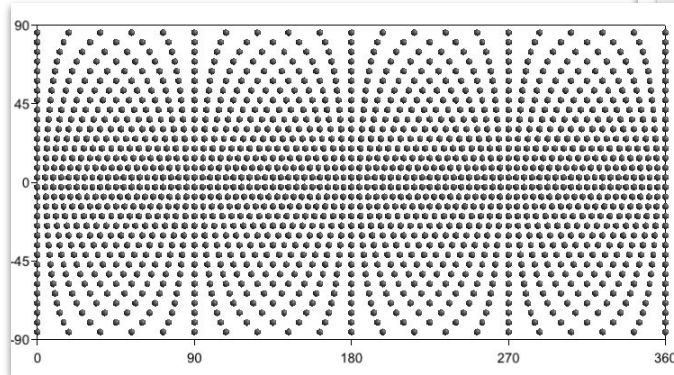
Octahedral Gaussian (O24) grid example from

<https://doi.org/10.5194/gmd-12-651-2019> (Fig. 2, cropped):

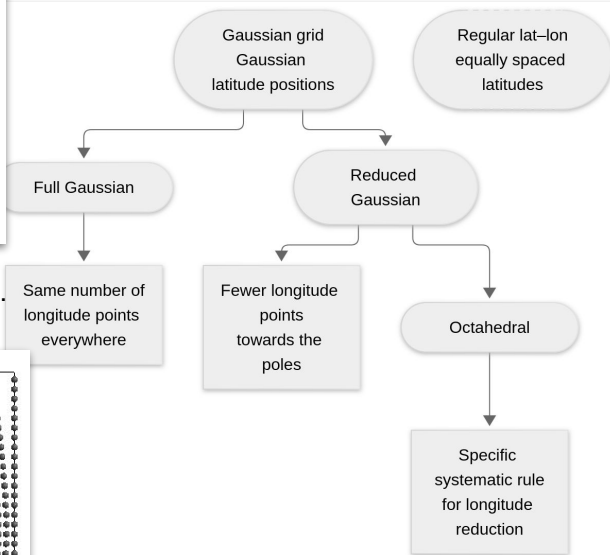


O16 grid example shown in projected lat vs. lon. form.

From: <https://sites.ecmwf.int/docs/atlas/design/grid/>:



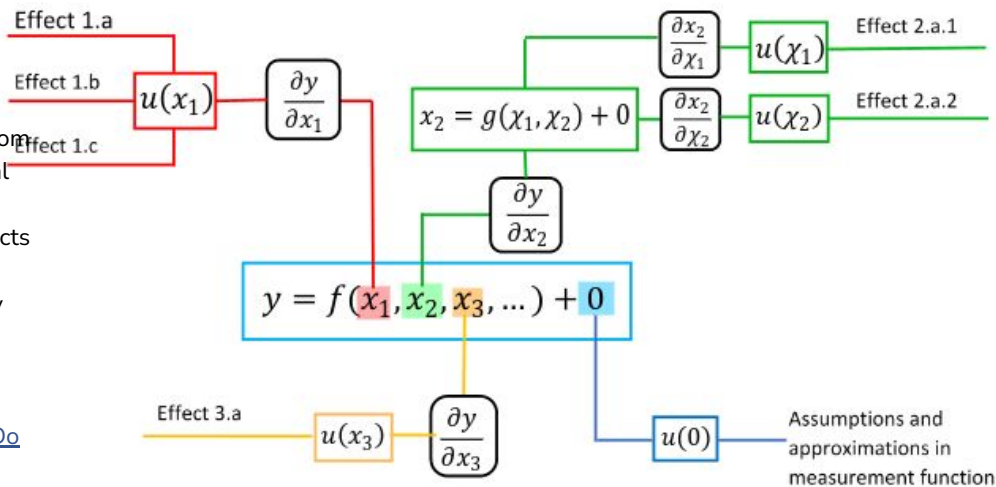
Schematic illustrating Gaussian grids e.g. how the 'reduced' case differs from 'full' case:



Major work in progress, unlikely to make it to CF 1.14

- Uncertainty metadata [[#660](#)]. As copied verbatim:
 - CF still needs to **formalise support for metadata that describes the uncertainty in the measurement of a quantity** described by a data variable
 - Proposed to do so with the introduction of a *new uncertainty variable*, which is a container for uncertainty information. It may be thought of as a special type of ancillary variable, in that it provides metadata which varies over the sampling domain of a data variable. An uncertainty variable has standardised attributes to describe the nature of the uncertainty, and its data array provides the actual uncertainty values.
 - The probability distribution of the quantity being measured (i.e. the measurand) can be described either not at all, very simply (e.g. just noting that it is Gaussian), or in great detail (e.g. by kurtosis parameter values which vary over the domain, and error-correlation coefficients between arbitrary dimensions)
 - Builds on work by Sam Hunt and Pieter de Vis towards the [Uncertainty Metadata Naming Conventions \(UNC\)](#)

Illustration of the potential complexity of uncertainty quantification via a 'Conceptual Uncertainty Tree Diagram', taken from 'General guidance on a metrological approach to fundamental data records (FDR), thematic data products (TDP) and fiducial reference measurements (FRM) – Uncertainty Analysis Process' by the UK's National Physical Laboratory (NPL)QA4EO Team, at https://qa4eo.org/docs/3_Process_Document.pdf:



Many thanks to everyone who contributed in any way to CF 1.14 (or earlier)! CF is a community effort and we rely community contributions*



*or will contribute still - you have a little time left!



In summary

- Since 2023 we've made **yearly releases a few months after each annual workshop**
- **CF 1.14** is this year's imminent version, replacing Dec 2025 version 1.13
- Changes *definitely* going into 1.14 (already merged ready for release) include:
 - Reduced Gaussian grid support [[#619](#)]
 - A new parametric vertical coordinate, 'atmosphere hybrid sigma log-pressure coordinate' [[#647](#)]
 - Clarify units prohibition of numerical factors and offsets [[#620](#)]
 - Section '4.3. Vertical (Height or Depth) Coordinate' clarity improvements [[#635](#)]
 - Formatting/consistency and infrastructure improvements
- Standardisation of uncertainty metadata [[#660](#)] is a major endeavour in progress but is unlikely to make it into v1.14

